
Group 4

**SmartSpend AI
Master Test Plan**

Version 1.1

SmartSpend AI	Version: 1.1
Master Test Plan	Date: 09/08/26
SS-MTP-01	

Revision History

Date	Version	Description	Author
17/07/22026	1.0	Draft	Phan Văn Bá Đạt
09/08/2026	1.1	Updated with complete functional components, use-case mapping, and test coverage	Phan Văn Bá Đạt

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Master Test Plan

1. Introduction

This Master Test Plan document establishes a comprehensive framework for validating the functional components and cross-platform workflows of the **SmartSpend AI** application.

- **What are the objectives of this document?**
To define the precise testing scope, establish validation standards for our core AI/OCR integrations, optimize resource allocation, and identify environmental needs across mobile client testing matrices.
- **What is this document about?**
It details the systematic approach to testing the application's primary functional pipelines, including automatic metadata extraction from expense documents, local manual inputs, custom system parameters, and cross-platform profile infrastructure.
- **Who should use it?**
This document is intended for project developers executing unit/integration fixes, student testers conducting manual behavioral validation, and the course instruction team assessing the technical implementation criteria.
- **What are the key features of this document?**
Key features include a targeted functional layout mapping AI-driven and manual workflows, modern mobile-first environment criteria, and dynamic team role assignments adapted to agile project constraints.

1.1. Purpose

The purpose of this document is to ensure that the core operational components of the SmartSpend AI platform meet structural software engineering expectations before deployment, mitigates data anomalies in local accounting systems, and guarantees consistency across heterogeneous mobile runtimes.

2. Target Test Items

2.1 Functional Testing

Module 1: Authentication & User Profile Management

- **UC01: Register:** Validates new account creation via email/password (with OTP verification) and Google Sign-In, input sanitization for name/email/password, OTP expiration handling, and initial profile setup.
- **UC02: Login:** Validates user authentication credentials (email/password and Google login), error handling for incorrect inputs, "Forgot Password" redirection, and Multi-Factor Authentication (MFA) OTP flows.
- **UC03: Forgot Password:** Validates password recovery via email OTP verification, error handling for unrecognized emails, OTP timeout management, and new password security rule enforcement.
- **UC04: Manage Profile:** Validates personal information updates (name, birthday), profile picture uploads via camera/gallery, and permission handling (Gallery/Camera access permission allow/deny flows).
- **UC05: Change Password:** Validates authenticated password modification by verifying old password accuracy and enforcing new password strength constraints.

Module 2: Transaction & Category Management

- **UC06: Manage Custom Categories:** Validates CRUD operations (Create, Read, Update, Delete) on custom spending categories, unique name validation, icon/color customization, and deletion re-mapping to 'Uncategorized'.
- **UC07: Add Manual Transaction:** Validates manual entry of expenses/incomes (Name, Type, Amount, Date/Time, Category, Note), field validation for amounts > 0, and immediate ledger synchronization.
- **UC08: Manage Transactions:** Validates editing and deleting existing transaction records, balance recalculations, and confirmation prompts.
- **UC09: View Transaction History:** Validates searching, filtering (by date, type, amount, category), and scrolling through historical transaction logs.

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Module 3: Budgeting & Financial Reports

- **UC10: Set Monthly Budget:** Validates setting, updating, and saving monthly budget targets per category or overall account.
- **UC11: Receive Budget Warning:** Validates system alert generation and notification triggers when spending thresholds approach or exceed allocated budgets.
- **UC12: View Financial Reports:** Validates aggregation of transaction data into visual summary charts and verification of financial report exports (Excel/CSV).

Module 4: AI Receipt Scanner

- **UC13: Add Transaction via AI Scanner:** Validates image capture/upload of expense receipts, OCR text extraction (Total Amount, Date, Merchant Name), AI automated categorization, and review/save workflow.

Module 5: Administration & System Management

- **UC14: Manage System Dashboard:** Validates web-based administrative dashboard metrics, overview panels, and system health status.
- **UC15: Manage User Accounts:** Validates admin oversight of registered user accounts, status updates, and account moderation.
- **UC16: Manage Push Notifications:** Validates configuration and dispatch of system push notifications to users.
- **UC17: View AI Scan Logs:** Validates monitoring, querying, and auditing of AI receipt scanning processing logs and API calls.
- **UC18: Manage User Feedback:** Validates submission of user feedback from the mobile client and administrative review/management of feedback logs on the dashboard.

2.2 Usability Testing

- **Mobile Interface Ergonomics:** Validation of touchscreen hit-box mapping, typography readability on various screen sizes, and proper scaling on different device resolutions.
- **Workflow Efficiency:** Verification that the critical path for AI receipt scanning and automated expense generation executes seamlessly within 3 active application screen interactions.
- **User Error Prevention:** Evaluation of contextual error dialogs, system alerts, and clear confirmation cues when users attempt destructive data actions (e.g., deleting a transaction or custom category).

2.3 Performance Testing

- **AI/OCR Processing Latency:** Benchmarking the turnaround duration from the execution of the image upload post-request to the population of extracted text fields on the UI, ensuring it does not trigger application freezes or OS ANR (Application Not Responding) dialogs.
- **Local UI Responsiveness & Frame Rates:** Measuring interface smoothness (maintaining optimal frames per second) during continuous scrolling inside highly populated transaction logs and dynamic chart rendering.

2.4 Security Testing

- **Data Transport Encryption:** Verification that all client-server communications, API requests containing user financial metadata, and image data streams are transmitted exclusively over secure cryptographic protocols (HTTPS/TLS).
- **Session Security and Storage:** Validation of secure token storage mechanisms on the local mobile client (e.g., using Android Keystore / iOS Keychain) to prevent unauthorized token exposure.
- **Input Validation & Form Injection Prevention:** Testing structural field sanitization across manual input fields to ensure malicious string entries or scripts are neutralized before processing.

3. Environmental Needs

3.1 Hardware Requirements

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Hardware	Requirement
Android Mobile Device	Android smartphone with camera support
IOS Mobile Device	IOS smartphone with camera support
Internet	Stable Wi-Fi or mobile network connection
Development Computer	Windows 11 or macOS with Android Studio installed

3.2 Software in the Test Environment

The following base software elements are required in the test environment for this *Test Plan*.

Software Name	Purpose	Version	Type and Other Notes
Android OS	Mobile operating system	Android 11 or later	Primary testing platform on physical devices to validate camera workflows and native behavior
IOS	Mobile operating system	IOS 15 or later	Secondary testing platform for cross-platform visual layout and compatibility validation
Android Studio	Emulator and development environment	Latest Version	Used for local execution, log inspection, and emulator testing
IOS Simulator	Client layout and structural rendering verification	Included with Xcode	Integrated native emulation platform running within Apple Xcode architectures
Google Chrome	Backend/Admin testing	Latest Version	Standard web browser used to access management dashboards and backend configurations
Supabase	Authentication and database services	Latest Version	Cloud backend service hosting user session data and relational financial tables.
OCR Engine (Google ML Kit / Tesseract / API used by project)	Receipt text recognition and metadata extraction	Latest Version	Core AI OCR component responsible for parsing raw image arrays into string fields

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Software Name	Purpose	Version	Type and Other Notes
Backend API	Business logic and database access	Current project version	Test server processing transactional compute operations before DB commits

3.3 Productivity and Support Tools

The following tools will be employed to support the test process for this *Test Plan*.

Tool Category or Type	Tool Brand Name	Vendor or In-house	Version
Defect & Task Tracking	Jira Software	Vendor	Current Web Interface
Version Control Infrastructure	GitHub Repositories	Vendor	Current Distribution
API Testing & Automation	Custom Test Scripts (VS Code / Python)	In-house	Latest Version
Database & Auth Management	Supabase Dashboard	Vendor	Current Cloud Version
Log Inspection & Debugging	Android Logcat / Xcode Console	Vendor	Integrated with IDEs

4. Responsibilities, Staffing, and Training Needs

4.1 People and Roles

This table shows the staffing assumptions for the test effort.

Human Resources		
Role	Minimum Resources Recommended (number of full-time roles allocated)	Specific Responsibilities or Comments
Test Manager	1 (Phan Văn Bá Đạt)	Provides management oversight. Responsibilities include: - planning and logistics - agree mission - identify motivators - acquire appropriate resources - present management reporting - advocate the interests of test - evaluate effectiveness of test effort

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Human Resources		
Role	Minimum Resources Recommended (number of full-time roles allocated)	Specific Responsibilities or Comments
Tester	1 (Phan Văn Bá Đạt)	Implements and executes the tests. Responsibilities include: • implement tests and test suites • execute test suites • log results • analyze and recover from test failures • document incidents
Test System Administrator	1 (Phan Văn Bá Đạt)	Ensures test environment and assets are managed and maintained. Responsibilities include: • administer test management system • install and support access to, and recovery of, test environment configurations and test labs